

Noor Aldeen Almusleh

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An AI/ML engineer and researcher with over four years of experience worldwide, across academia and the industry. Experienced with project development from early development through to production, and with different technologies, including state of the art machine learning (ML) algorithms, computer vision (CV), computer graphics (CG), and reinforcement learning (RL). Demonstrated the ability to understand and organise customer needs, and meet the stakeholders' targets and deadlines. Displays a positive outlook on challenges and is able to manage priorities.

Education

M.Sc. in Artificial Intelligence (Distinction)

University of Bath, United Kingdom

(May 2026)

- Postgraduate programme focused on machine learning, deep learning, computer vision, graph neural networks, and RL, with supporting modules in optimisation and mathematical foundations for AI.
- Thesis: “**Reinforcement Learning for 2D Character Line-Art Generation.**” Designed a pipeline in which an RL agent trained with *Proximal Policy Optimisation (PPO)* learns to render character strokes sequentially. The agent utilises a *GNN*-based encoder to represent the agent's observation.
- Developed a generative model based on *Variational Autoencoder (VAE)*, providing compact supervisory signals and reducing RL sample complexity; integrated the VAE with the RL agent to accelerate convergence and improve sample quality.

B.Sc. in Physics (First-Class Honours)

Al Al-Bayt University, Jordan

(Jan 2018)

- Four-year programme (uses English as the language for instruction) with rigorous training in astrophysics, optics, mathematical physics, quantum mechanics, electromagnetism, statistical mechanics, and thermodynamics.
- Computational Materials Modelling (DFT/VASP): Performed DFT calculations in VASP to obtain structure- and energy-level data, and used these simulations to generate features for ML models to accelerate materials screening beyond pure DFT workflows.
- Developed an early foundation in scientific computing, numerical optimisation, and modelling, which was later extended to AI-for-science projects (perovskites, SOFC materials).
- Dissertation: “Heavy Quarkonium Mass Spectra from Complex Potentials.” Solved the Schrödinger equation for heavy mesons (e.g. bottomonium) under complex potentials using numerical methods, demonstrating problem formulation, model selection, and result validation.

Experience

Graduate Researcher

2024 – Present

University of Bath, United Kingdom

- Planned and organised a project to create an RL agent that can imitate drawing.
- Met project goals by researching algorithms, designing and developing code for a drawing environment, a deep RL framework based on PPO, and a novel generative model architecture.
- Utilised different sources of data, including image and graph data and processed rasterised images with Python OpenCV2, while graph data was processed with frameworks such as NetworkX, PyTorch Geometric, and Python Optimal Transport.
- Developed a state-of-the-art generative model architecture and trained a deep RL agent (PPO) to select drawing actions; used a GNN to encode the structural graph and CNN to encode image data.

- Built models using PyTorch and optimised performance with PyTorch profiling for code optimisation to fit the models on limited computational resources (1–2 GPUs).

AI Engineer (Scientific Discovery) Korea Energy Research Institute, South Korea

2023 – Present

- Led cross-functional AI adoption across multiple research teams by defining ML use cases and translating scientific/energy-engineering problems into data-driven models and frameworks.
- Conduct applied research in AI for scientific discovery and materials science, collaborating with domain experts across the globe to develop publishable results; contribute to and co-author peer-reviewed papers in Q1-ranked journals, supporting analysis, visualisation, technical writing, and responses to reviewers.
- Understood and organised investors' needs to develop and improve analytics models using proprietary data, processed using scientific Python stack (e.g. NumPy, Pandas, Scikit-Learn, and SciPy)
- Developed physics-informed neural networks using statistical learning and deep learning techniques, and version control via Git and GitHub to track, document, and collaborate on code with team members.
- Developed novel AI models and end-to-to-end pipelines to interpret the electrical behaviour of perovskites, presenting results through tailored visualisations including interactive dashboards and Streamlit applications.

AP Physics Lead Teacher Ministry of Education, Dubai, United Arab Emirates

2019 – 2023

- Managed and mentored a team of four Physics teachers, coordinating curriculum planning, assessment design, moderation, and professional development; maintained consistent instructional quality across classes and aligned delivery with Ministry requirements and AP standards.

Research Papers

- Graph–Convolutional Variational Autoencoder for Character Line Art
 - N. A. Almusleh, J. Derungs, W. Li
 - In preparation
 - Unleashing Sodium–Sulfur Battery Performance With Atomically Dispersed Single Atom Catalysts
 - S. Maiti, M. Curnan, N. Almusleh, S. Subhalaxmi, D. Shin, R. Narayan, K. Maiti, J. Hur
 - DOI: <https://doi.org/10.1002/aenm.202505686>
 - Advanced Energy Materials (Accepted: January 2026)
 - Metal-poor Stars Observed with the Automated Planet Finder Telescope. III. CEMP-no Stars are the Descendants of Population III Stars
 - N. A. Almusleh, A. Taani, S. Özdemir, M. Rah, M. A. Al-Wardat, G. Zhao, M. K. Mardini
 - DOI: <http://doi.org/10.1002/asna.202113867>
 - Astronomische Nachrichten (Accepted: June 2021)
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Patents

- AI-Based Framework for Diagnosing and Optimising Perovskite Solar Cell Performance Using Physics-Informed and Generative Models
 - Inventor: N. Almusleh et al., USPTO (US), Patent application filed, Application No. 10-2025-0200064, 2026